Exam Room Allocation using Integer Optimization

1 Introduction

This document describes an integer programming model for assigning exams (each row is a exam-student-course record) to room-time slots subject to capacity, scheduling, and special-accommodation constraints. The model enforces that exams that share a room slot are either *concurrent* (identical start time) or *non-overlapped and separated* by at least 15 minutes, and it prefers 30-minute pre- and inter-exam buffers via a lexicographic objective.

2 Sets and Indices

- $i \in E$: set of exam-student-course records
- $j \in R$: set of room-time slots
- $c \in C$: set of courses
- $A \subseteq E \times R$: compatible (exam, room-slot) pairs.
- $R_i = \{\, j : (i, j) \in A \,\}$: rooms compatible with exam i.
- $E_j = \{\, i : (i, j) \in A \,\}$: exams compatible with room j.
- $G_{j,t} = \{\, i \in E_j : start_i = t \,\}$: the *concurrent exam-group* in room j with start time t.

3 Parameters

3.1 Exam Parameters

- course_i: course of exam i
- $\text{RD}_i \in \{0, 1\}$: reduced distraction requirement
- $\text{PRIV}_i \in \{0, 1\}$: private room requirement

3.2 Room Parameters

- cap_j: capacity of room j
- $\text{zone}_j \in \{1, 2\}$: zone of room j

3.3 Compatibility

$$(i,j) \in A \iff \begin{cases} \text{date}_i = \text{date}_j \\ \text{start}_i \geq \text{start}_j + 15 \\ \text{end}_i \leq \text{end}_j \end{cases}$$

3.4 Buffer and gap sets

Let

$$B^{30} = \{(i,j) \in A : start_i - 30 \geq start_j\}, \qquad \text{(30-min pre-buffer)}$$
$$B^{15+} = \{(i,j) \in A : end_j \geq end_i + 15\}. \qquad \text{(15-min post-buffer)}$$

For every room slot j and every pair of non-concurrent exams $i_1, i_2 \in E_j$ (i.e. $s_{i_1} \neq s_{i_2}$), define the inter-exam gap $\Delta(i_1, i_2) = \max(start_{i_2} - end_{i_1},\ start_{i_1} - end_{i_2})$ and the sets

$$H_j = \{(i_1, i_2) : \Delta(i_1, i_2) < 15\} \qquad \text{(hard conflict – forbidden)}$$
$$Q_j = \{(i_1, i_2) : 15 \leq \Delta(i_1, i_2) < 30\}. \qquad \text{(allowed but penalised)}$$

4 Decision Variables

$$x_{i,j} = \begin{cases} 1 & \text{if exam } i \text{ assigned to room } j \\ 0 & \text{otherwise} \end{cases}$$

$$y_j = \begin{cases} 1 & \text{if room } j \text{ is used} \\ 0 & \text{otherwise} \end{cases}$$

$$z_{c,j,t} = \begin{cases} 1 & \text{if course } c \text{ presents in group } G_{j,t} \\ 0 & \text{otherwise} \end{cases}$$

$$r_{j,t} = \begin{cases} 1 & \text{if an RD student is present in group } G_{j,t} \\ 0 & \text{otherwise} \end{cases}$$

$$q_{i_1,i_2,j} = \begin{cases} 1 & \text{if both exams } i_1,\ i_2 \text{ assigned to room } j \text{ with only a 15-min gap} \\ 0 & \text{otherwise} \end{cases}$$

5 Constraints

C1 – Unique assignment

$$\sum_{j \in R_i} x_{ij} \leq 1 \quad \forall i \in E$$

Room Slot Activation

$$x_{ij} \leq y_j \quad \forall (i,j) \in A$$

C2 – Capacity constraint, per exam-group

$$\sum_{i \in G_{j,t}} x_{ij} \leq \text{cap}_j \quad \forall j \in R$$

Course Linking For every $j \in R$, $c \in C$ with $G^c_{j,t} := \{i \in G_{j,t} : \text{course}_i = c\} \neq \emptyset$:

$$\begin{aligned} z_{c,j,t} &\geq x_{ij} && \forall\, i \in G^c_{j,t}, \\ z_{c,j,t} &\leq \sum_{i \in G^c_{j,t}} x_{ij}, \end{aligned}$$

C3 – At most three courses per concurrent group

$$\sum_{c \in C} z_{c,j,t} \leq 3 \quad \forall j \in R$$

C4 – Reduced-distraction cap

$$\begin{aligned} r_{j,t} \geq x_{ij} \quad \forall i \in G_{j,t} \text{ with } \mathrm{RD}_i = 1 \\ \sum_{i \in G_{j,t}} x_{ij} \leq 20 r_{j,t} + \mathrm{cap}_j (1 - r_{j,t}) \end{aligned}$$

C5 – PRIV / CODS private room

$$\sum_{k \neq i,\, k \in G_{j,start_i}} x_{kj} \;\leq\; \mathrm{cap}_j\,(1 - x_{ij}) \quad \forall i \in G_{j,t} \text{ with } \mathrm{PRIV}_i = 1$$

C6 – Pre-exam buffer (hard) Enforced implicitly in the definition of A. The 30-minute preference is handled in the objective through B^{30}.

C7 – Inter-exam gap For every slot j:

$$\begin{aligned} x_{i_1,j} + x_{i_2,j} &\leq 1 && \forall\,(i_1, i_2) \in H_j, \quad \text{(hard: gap < 15 min forbidden)} \\ q_{i_1,i_2,j} &\geq x_{i_1,j} + x_{i_2,j} - 1 && \forall\,(i_1, i_2) \in Q_j. \quad \text{(soft: link for P2 penalty)} \end{aligned}$$

6 Objective Function

We solve a lexicographic multi-objective optimization:

P5) $\max \sum_{(i,j) \in A} x_{ij}$ — assign as many exams as possible

P4) $\min \sum_{j \in R} y_j$ — use as few room slots as possible

P3) $\min \sum_{(c,j,t)} z_{c,j,t}$ — concentrate each course into as few exam groups as possible

P2) $\min \sum_{(i,j) \in A \setminus B^{30}} x_{ij} + \sum_{(i_1,i_2) \in Q_j} q_{i_1,i_2,j}$ — prefer 30-min pre- and inter-exam buffers

P1) $\min \sum_{(i,j) \in A \setminus B^{15+}} x_{ij}$ — prefer $\geq$ 15-min post-exam buffer

P0) $\min \sum_{(i,j) \in A, \mathrm{zone}_j \neq 1} x_{ij}$ — prefer Zone-1 rooms over Zone-2

7 Conclusion

The model ensures feasible and efficient allocation of exams to rooms while respecting operational constraints and preferences.